Rohan Garg

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Education

University School of Information, Communication and Technology, Delhi *July 2025 - Jun 2029*
Bachelor of Technology in Information Technology ***CGPA: 8.3***

Darbari Lal DAV Model School, New Delhi *2023 - 2025*
Class X: ***93%*** | *Class XII:* ***89%***

Technical Skills

Languages: C, C++, JavaScript, TypeScript, Python, SQL, HTML5/CSS3
Backend & Databases: Node.js, Express.js, Next.js (App Router), PostgreSQL, Prisma ORM, Redis, MongoDB, Supabase
Architecture & Protocols: RESTful APIs, WebSockets, WebRTC (LiveKit), Microservices, System Design, RBAC
Tools & Infrastructure: Git, Docker, Render, Postman, Vercel, Tailwind CSS, Linux/Bash, VS Code
Core Coursework: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks

Work Experience

Association for Computing Machinery - ICPC *Oct 2025 - Jun 2026*
Club Coordinator *New Delhi*

- Conducted an **8-session DSA Bootcamp**, mentoring 100+ students through **75+ curated algorithmic problems**.
- Maintained ACM chapter backend services and engineered automated event and programming challenge management scripts.

IEEE *Oct 2025 - Jun 2026*
Tech Team Member *New Delhi*

- Engineered high-throughput **RESTful APIs**, normalized database schemas, and backend microservices using Node.js and MongoDB for 3+ university applications.
- Coordinated infrastructure execution for a regional hackathon with AIC GGSIPU, supporting 100+ teams end-to-end.

Projects

CollabSphere | **Real-Time Messaging Platform** | *Next.js, Redis, PostgreSQL, Prisma, WebSockets, LiveKit* | Live | Code

- Engineered a real-time messaging engine using **WebSockets** and WebRTC audio/video streaming powered by **LiveKit**.
- Built an automated background cleanup worker and disappearing message system using **Redis TTL eviction**, drastically reducing PostgreSQL clutter.
- Architected a scalable relational schema with **Prisma ORM** and **PostgreSQL**, incorporating fine-grained RBAC and anonymous channels.
- Optimized frontend delivery with Next.js Suspense skeleton loaders, deploying the production stack containerized on **Render**.

JugaadHub | **Campus Peer-to-Peer Rental Platform** | *Next.js, Supabase, PostgreSQL, Razorpay, Cloudinary* | Live | Code

- Architected secure transactional workflows including **escrow deposits, wallet management, and Razorpay payments** for peer-to-peer campus rentals.
- Implemented real-time messaging, OTP handover verification, and KYC workflows utilizing **Supabase RPC functions and PostgreSQL**.

VideoTube | **Full-Stack Video Engine** | *Node.js, Express.js, MongoDB, React, JWT, Cloudinary* | Live | Code

- Designed RESTful APIs for video uploading, media processing, subscriptions, and playlists using **JWT authentication** and session handling.
- Optimized complex relational data retrieval and aggregation performance using **MongoDB Aggregation Pipelines**.

Achievements & Competitive Programming

- Solved **450+ data structures and algorithmic problems on LeetCode**, demonstrating strong core problem-solving proficiency.
- Cleared **two rounds of the Google Big Code Challenge 2026**, demonstrating strong analytical skills.
- Achieved **Top 100 International Rank** in the International Mathematics Olympiad (IMO) with multiple 1st-place regional finishes.
- Selected for **Round 2 at Smart India Hackathon**, placed Top 10 at HackXtract, finalist at HackNite at and secured **Top 7 at Orbix Hackathon, IIIT Delhi**.